

upgrades and even new equipment in the market.

Dense wavelength-division multiplexing tester. Data-hungry applications like video-streaming demand increased bandwidth. Thus, dense wavelength-division multiplexing (DWDM) technology is gaining popularity in the market because it multiplies the data carrying capacity of optical fibre by 72 times using C band. The number of channels increases beyond 100 using L and S bands. Before DWDM came into existence, wavelength-division multiplexing (WDM) and passive wavelength-division multiplexing (PWDM) were used.

Now, DWDM has made its way into testing of metropolitan area network (MAN) and mobile back-haul applications. Experts explain that a single fibre can carry several hundreds of wavelengths. DWDM combines and then transmits multiple signals simultaneously at different wavelengths in a fibre. This increases the physical capacity of a medium. DWDM instruments thus smoothen the working of and services provided by optical fibres.

Internet protocol tester. This is a new concept in the fibre optics field. As optical fibres are laid out over long distances, there is a high probability of data losses or errors in data packets, which may delay the process. If there is no testing for data efficiency, the fibre optics would be an expensive loss to customers. It is like having a LAN setup in a place where the user is not able to avail all the services smoothly due to the poor fibre optics connection or quality. Manufacturers are using IP tools to measure the 100Gbps speed. While in mobile applications, the speed has increased from existing 2Mbps to 10Gbps.

IP testers relay exact issues and deformities in a network. These check networks mainly for



EXFO FTB-8810NGE 100GB Ethernet tester



EXFO FTB-5240S DWDM tester



MT9083B2 OTDR tester from Anritsu India

accuracy, latency and bandwidth. Basically, IP testers check the stress and reliability of an optical fibre due to high-speed data travel and network, respectively. These tests ensure minimal interruptions during voice-over calls or video chats.

As optical fibres can support speeds up to 100Gbps, these can be stretched to major cities and long distances. Using these test equipment has become all the more crucial as the government has now come up with a set of protocols that follow international standards related to quality of optical fibre, interoperability issues and, above all, upgrading the speed of data carrier packets.

Optical time-domain reflectometer. Tripathi emphasises on the usability of optical time-domain reflectometers (OTDRs), which are used to detect faults or break-points in optical fibres. Over time, OTDRs have become compact and handheld devices for multiple applications. There is also a change in the source, display and control panel. Earlier, OTDRs used to support dual wavelength, but now these support four wavelengths at a time. Tier 2 fibre certification requires the use of OTDR for testing the performance and quality of the fibre length and linked components like cables, connectors and splices.

There is also an upgrade in OTDR operation as it now supports the cloud. So engineers can locate faults in the fibre without any hassle or wastage of time.

Dhawan cites a project where a centralised monitoring system gave the details of fibre health condition during Commonwealth Games in Delhi. This was done at Network Operation Centre (NOC), where digital map of Delhi was displayed on a big screen. Any breakage or degradation was graphically displayed on the map as well as coordinated through GPS, reducing the downtime of the networking.

Dhawan claims that some giant service providers also have centralised monitoring systems in place.

Optical spectrum analyser. Optical spectrum analysers are used during optical cable production and design. These have a spectral resolution of around 312MHz and perform scanning with full resolution in a second's time. These are connected using the USB or Ethernet ports.

To sum up

While wireless communication has become the buzzword, wireline technology continues to play a pivotal part, thus giving wireline equipment market plenty of scope to evolve and innovate. **EFY**